



LEMON



Visvesvaraya Trade Promotion Centre (VTPC)

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Preface

With a robust agro potential the Country has, the Government of India has launched the One District One Focus Product Scheme (ODOFP) for agriculture sector by the Ministry of Food Processing Industries. Among its primary objectives, the ODOFP initiative aims to enhance the value of the products which is eventually expected to boost employment and income levels of the farmers. The identified products across the country is available at <https://www.nfsm.gov.in/odopstatecropsreport.aspx>

Karnataka has identified its product mix with high demand and export potential across all the districts. The implementing agency for this initiative is the Karnataka State Agricultural Produce Processing and Export Corporation Limited (KAPPEC), the nodal agency in the State.

VTPC being the nodal agency for promotion of exports from the State has made an attempt to draw the action plan district wise, to capture the potential, present status and future prospects in domestic and international markets. Besides, detailed insights have been provided into the biological description of the product, their local, national and international varieties, export-import analysis, HS codes. For a holistic perspective for those concerned, each report also provides SPS standards, processing technologies available, export grading and packing specifications, and suggested pivotal roles and responsibilities among the government departments, boards, corporations and Universities.

Presenting the Action Plan/Report for Lemon, a ODOFP product mapped to Vijayapura district of Karnataka, formulated by VTPC. I sincerely hope that this ready reckoner with first-hand information regarding the farming sector would help all those concerned, especially the FPOs and food processing entrepreneurs who have a desire to focus on exporting this product.

S.R. Satheesha

Director (Exports) & Managing Director

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Introduction

Citrus lemon or 'Nimbu,' as they are called in India, is the most important fruit all over the world and is famous for its refreshing and sour flavour and used in culinary preparations. It is well known for nutritional as well as its medicinal property, both in traditional and modern medicine. The pulp and rind are also used in cooking and baking. The juice of the lemon comprises of about 5% to 6% citric acid, with a pH of around 2.2, giving it a sour taste. It is also rich in vitamin C and contains smaller amounts of Vitamin B, particularly thiamine, riboflavin, and niacin. Its juice is also used for non-culinary purposes like cleaning and stain removal.

Beginnings:

The lemon (*Citrus limon*) is a species of small evergreen tree in the flowering plant family Rutaceae, native to Asia, primarily Northeast India (Assam), Northern Myanmar or China. However there are speculations about the fruit being native to the state of Assam in India. Moreover, there is no difference between lemon and lime in India. All the varieties are referred to as Nimbu and the word Nimbu derives itself from the Persian word *līmūn*.

Citrus lemon Varieties famous across the World



1. Bonnie Brae

The 'Bonnie Brae' is oblong, smooth, thin-skinned and seedless. These are mostly grown in San Diego County, USA.

2. Lisbon

The Lisbon lemon is very similar to the Eureka and is the other common supermarket lemon. It is smoother than the Eureka, has thinner skin, and has fewer or no seeds. It generally produces more juice than the Eureka.



3.Eureka

The 'Eureka' grows year-round and abundantly. This is the common super market lemon, also known as 'Four Seasons' because of its ability to produce fruit and flowers together throughout the year. This variety is also available as a plant to domestic customers. There is also a pink fleshed Eureka lemon, with a green and yellow variegated outer skin. They are commercially popular because of their reliable flavour and continuous growing season.



4.Meyer

The 'Meyer' lemon is not 'true' lemon, but hybrid which has originated from China. It is a cross between a lemon and a sweet orange such as a mandarin. It looks like lime when young, rounder than true lemons and with a lime green skin. As the lemon ripens, it takes on the typical yellow shade, with a strong fragrance and thin skin. These lemons have a more subtle flavour than the Eureka or Lisbon lemons, which are widely available in grocery stores. Instead, they have a sweeter taste, with a dark yellow flesh and usually around 10 seeds.



5.Primofiori

Primofiori lemon is grown extensively in the Mediterranean region and is the most largely commercially produced lemon in Spain. The trees have a vigorous growth habit, with large leaves and dense foliage. The fruit it produces is pale yellow in colour, with a thin and smooth skin. Lemons can be round or oval and are smaller in size than most other lemon varieties, though they tend to be much juicier.



6.Verna

This lemon tree is native to Spain, where it is the second most important lemon tree after the Primofiori. It is widely known as both Verna and Berna. The fruits of this tree are less appealing for consumption than other lemons, as they tend to have a thick rind and do not contain much juice. This lemon is ornamentally attractive. The 'Sorrento' is native to Italy. This fruit's zest is high in lemon oils. It is the variety traditionally used in the making of limoncello (Italian lemon liquor). The 'Yen Ben' is an Australasian cultivar.

Lemon Cultivation in India & Availability

Common Names: Nimbu, Champra, Elumiccha The fruit has a variety of uses and is cultivated throughout the country. However, the Commercial production of lemons takes place in Andhra Pradesh, Maharashtra, Tamil Nadu, Gujarat, Rajasthan, and Bihar. Other states also cultivate it for their domestic use. As citrus fruits are less demanding and can grow in almost any soil, the lemons are available in the markets throughout the year. However, more in the North than in South India.

Top lemon producing states are Gujarat, Andhra Pradesh, Madhya Pradesh, Karnataka, Orissa and Maharashtra. Gujarat contributes 20 % share of total lemon production in India followed by Andhra Pradesh accounting for 18%, Madhya Pradesh and Karnataka accounting for 10% of total lemon production in India.

India is the largest producer of Lemons in the world. However, the size of the lemons is not as big as those of other countries. Nevertheless, it exports fresh lemons to countries like Japan, Thailand, Bangladesh, etc.

Lemon Varieties in India

There are many types of lemons that grow across the country, but we're covering the 8 most popular ones below:

1. Assam Nimbu/ Nemu Tenga

The name refers to the two types of lemons that grow in Assam, namely, Gol Nemu and Kazi Nemu. Gol Nemu is round and sweet. Further, it is usually larger than other varieties. GI tagged lemon of Assam Kazi Nemu, on the other hand, is juicier with an elongated oval shape. This Nemu is popular in Assam due to its use in Assamese cuisine.



2. Lisbon

This greenish-yellow, short-living variety is similar to Eureka (another lemon variety) in their elongated shape. The size of these lemons is average, with a prominent nipple at the apex. Apart from this, they are quite acidic in nature.



3. Sweet Lemon/Lime

This variety grows mainly in the regions of Malabar and Nilgiris. They are usually nonacidic in nature. You can eat this variety as a fruit rather than juicing it. These lemons or limes also go by the name of Mausambi in India.



4. Genoa

Commercial cultivation of this variety happens in India. These lemons are oval or oblong in shape and yellow in color, along with a decent amount of juice in them. Therefore, this variety of lemons is appropriate for pickles and marmalades.



5. Rough Lemon

This variety has such a name because of the rough, rugged, and irregular shape of its skin. It also bears a strong scent and flavor.

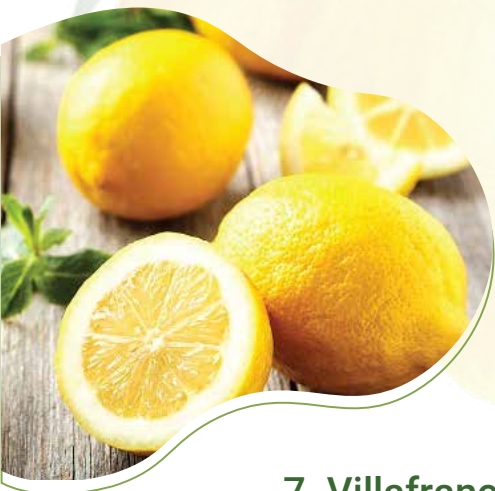
This variety is said to have its roots in the wilds of Northern India.

They have very less amount of juice and pulp.



6. Nepali Oblong/ Pat Nebu

The commercial cultivation of this variety takes place in India with its roots in Assam. This lemon variety has a smooth and glossy texture with medium acidity and very few or no seeds.



7. Villafranca

This variety of lemon is bright yellow in color with a noticeable nipple at the top. Moreover, just like that of Lisbon, it is also a short living and low yielding cultivar.



8. Nepali Round

A successful cultivar in the states of South India. Despite the name, the Nepali round variety also has its origins in India. This variety is very juicy and has very few seeds.



9. Kagzi Lemons

Rare variety of lemon largely grown in Indi Taluk of Vijayapura district of Karnataka. This variety will be the second one in securing Geographical indication (GI) tag after Manipur's Kachai Lemon. Kagzi has thinner rind, more juice and highest ascorbic acid content among all the lemon varieties known so far.



GI tagged Citrus fruits of India

Jalna Sweet orange
Maharashtra



Memong narang
Meghalaya



Kazi Nemu
Maharashtra



Tamenglong Orange
Manipur



Coorg Orange
Karnataka



Devanahalli Pomello
Karnataka



Nagpur orange
Maharashtra and
Madhya pradesh



Arunachal orange
Arunachal Pradesh



Khasi Madarin
Meghalaya



Kachai Lemon
Manipur

The primary plant compounds in lemons

1. Citric acid

The most abundant organic acid in lemons, citric acid may help prevent the formation of kidney stones.

2. Hesperidin

This antioxidant may strengthen your blood vessels and prevent atherosclerosis — the build-up of fatty deposits (plaque) inside your arteries.

3. Diosmin

An antioxidant used in some drugs that affect the circulatory system, it improves muscle tone and reduces chronic inflammation in your blood vessels.

4. Eriocitrin

This antioxidant is found in lemon peel and juice.

5. D-limonene

Found primarily in the peel, d-limonene is the main component of lemon essential oils and responsible for lemon's distinct aroma. In isolation, it can relieve heartburn and stomach reflux.

Lemon and its by-products

1. Lemon fruit (Existing)
2. Lime peel (Existing)
3. Lemon juice concentrate (Existing)
4. Lemon dehydrated powder
5. Lemon oil
6. Citric acid
7. Citrate of lime

Lemon products and its HS codes: APEDA

08055000	Lemons (Citrus Limon/Limonum) & Limes (Citrus Aurantifolia/Latifolia) Fresh / Dried
08059000	Other Citrus Fruit Fresh/Dried
20089912	Lemon Squash
22021020	Lemonade

World Export and Imports (2021)

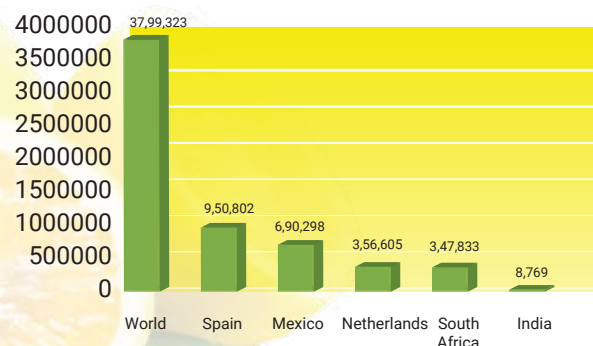
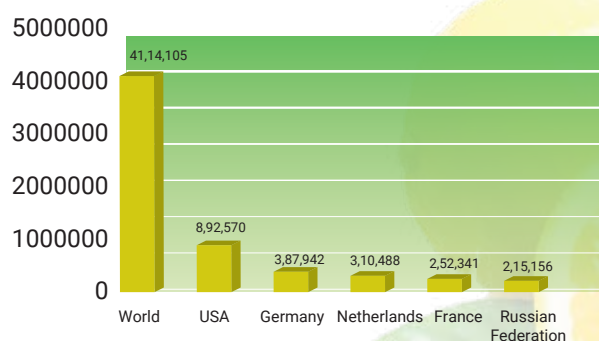
080550: Lemons (Citrus Limon/Limonum) & Limes (Citrus)

2021 Global Imports 4,114,105 USD.

2021 Global Exports 3,799,323 USD.

No	Country	Value (USD)
1	USA	892,570
2	Germany	387,942
3	Netherlands	310,488
4	France	252,341
5	Russian Federation	215,156

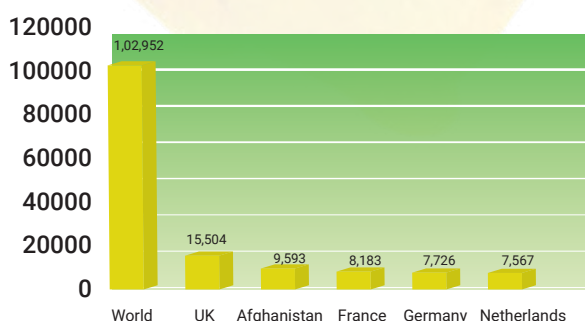
No	Country	Value (USD)
1	Spain	950,802
2	Netherlands	690,298
3	Mexico	356,606
4	South Africa	347,833
24	India	8,769



080590: Fresh or dried citrus fruit (excluding oranges, lemons "Citrus limon, Citrus limonum", limes "Citrus aurantifolia, Citrus latifolia", grapefruit, mandarins, incl. tangerines and satsumas, clementines, wilkings and similar citrus hybrids)

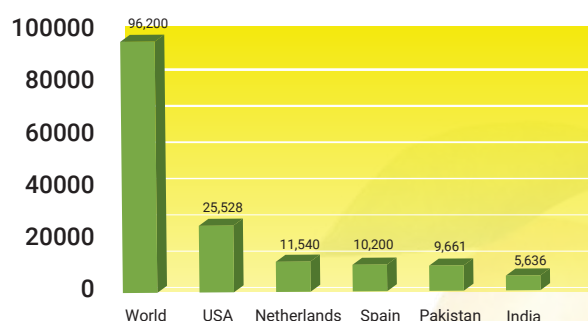
2021 Global Imports 102,952 USD.

No	Country	Value (USD)
1	United Kingdom	15,504
2	Afghanistan	9,593
3	France	8,183
4	Germany	7,726
5	Netherlands	7,567



2021 Global Exports 96,200 USD.

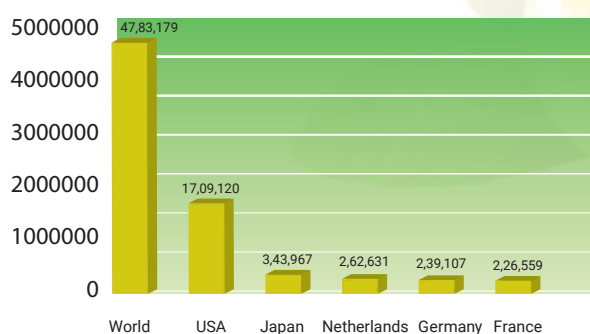
No	Country	Value (USD)
1	USA	25,528
2	Netherlands	11,540
3	Spain	10,200
4	Pakistan	9,661
24	India	5,636



200899: Fruit and other edible parts of plants, prepared or preserved, whether or not containing added sugar or other sweetening matter or spirit (excluding prepared or preserved with vinegar, preserved with sugar but not laid in syrup, jams, fruit jellies, marmalades, fruit purée and pastes, obtained by cooking, and nuts, groundnuts and other seeds, pineapples, citrus fruits, pears, apricots, cherries, peaches, strawberries, palm hearts and cranberries.

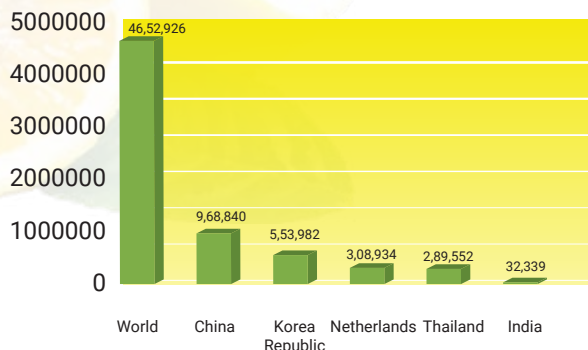
2021 Global Imports 4,783,179 USD.

No	Country	Value (USD)
1	USA	1,709,120
2	Japan	343,967
3	Netherlands	262,631
4	Germany	239,107
5	France	226,559



2021 Global Exports 4,652,926 USD.

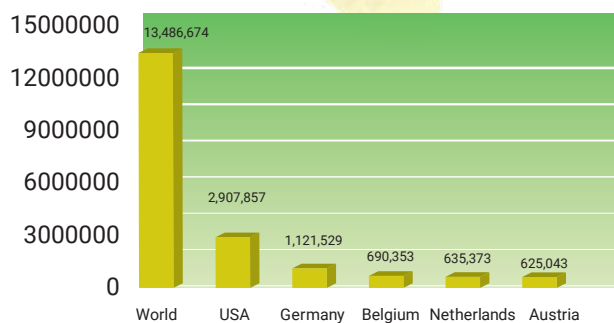
No	Country	Value (USD)
1	China	968,840
2	Korea Republic	553,982
3	Netherlands	308,934
4	Thailand	289,552
24	India	32,339



220210: Waters, incl. mineral and aerated, with added sugar, sweetener or flavour, for direct consumption as a beverage.

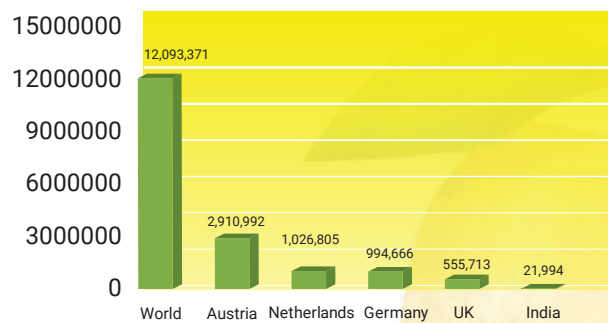
2021 Global Imports 13,486,674 USD.

No	Country	Value (USD)
1	USA	2,907,857
2	Germany	1,121,529
3	Belgium	690,353
4	Netherlands	635,373
5	Austria	625,043



2021 Global Exports 12,093,371 USD.

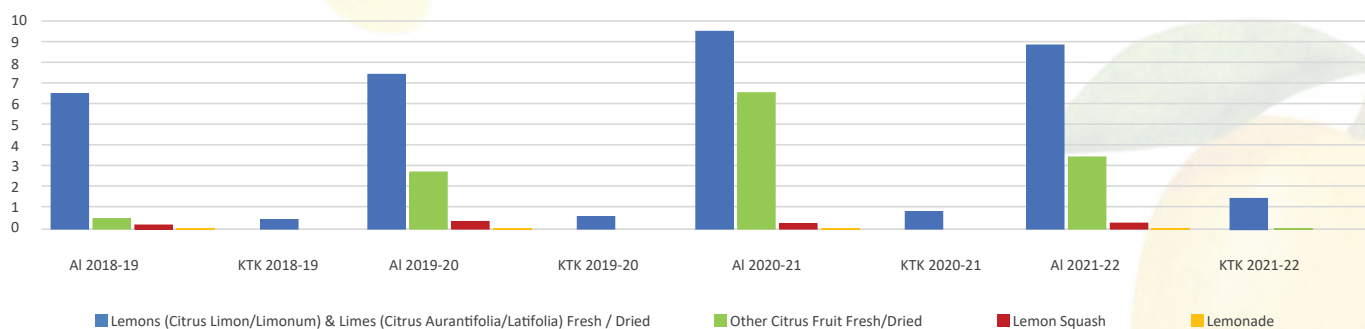
No	Country	Value (USD)
1	Austria	2,910,992
2	Netherlands	1,026,805
3	Germany	994,666
4	United Kingdom	555,713
24	India	21,994



Lemon Export Performance of last 4 years value in Mn USD

No.	HS Code	Product description	FY 2018-19		FY 2019-20		FY 2020-21		FY 2021-22	
			AI	KTK	AI	KTK	AI	KTK	AI	KTK
1	08055000	Lemons (Citrus Limon/Limonum) & Limes (Citrus Aurantifolia/Latifolia) Fresh / Dried	6.43	0.54 (3)	7.44	0.58 (3)	9.40	0.83 (3)	8.93	1.32 (3)
2.	0805900	Other Citrus Fruit Fresh/Dried	0.5	0.0 (6)	2.84	0.0 (7)	6.63	0.0 (15)	3.49	0.05 (6)
3	20089912	Lemon Squash	0.2	Nil (0)	0.34	0.0 (12)	0.23	0.0 (9)	0.26	Nil (0)
4.	22021020	Lemonade	0.04	Nil (0)	0.01	Nil (0)	0.01	Nil (0)	0.10	Nil (0)
5	29181400	Citric Acid	6.43	0.54 (3)	7.44	0.58 (3)	9.40	0.83 (3)	8.93	1.32 (3)

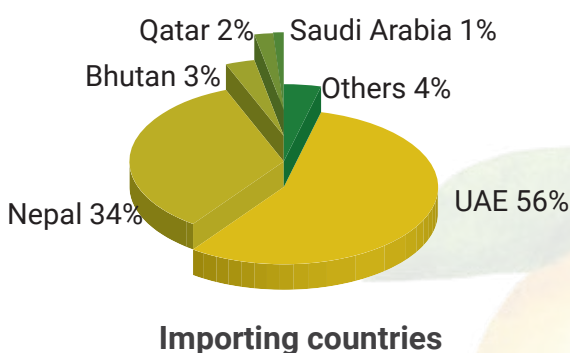
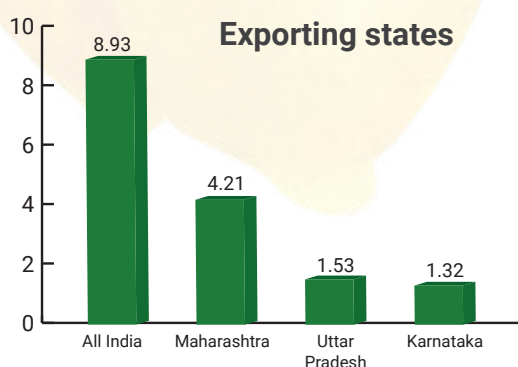
* The numbers in green indicate Karnataka rank in that year among other states.



Export Analysis of Lemon and its products : 2021-22

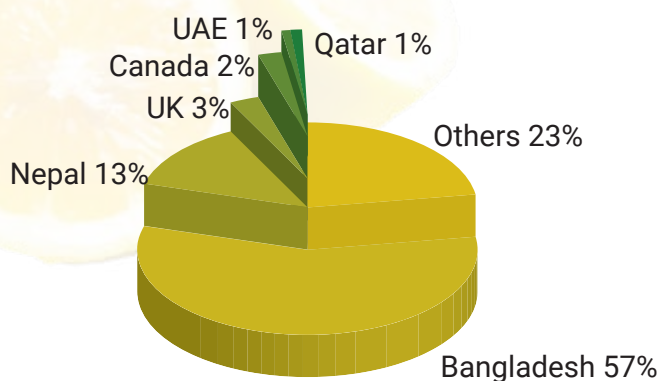
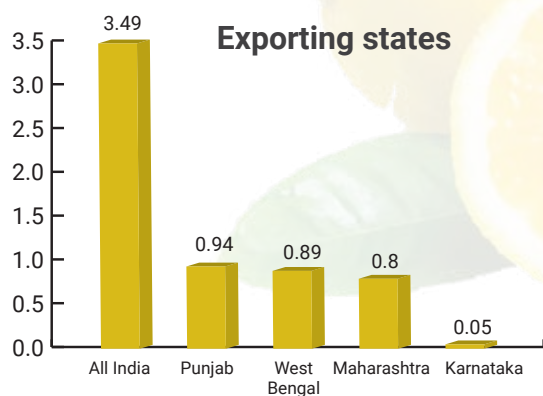
08055000: Lemons (Citrus Limon/Limonum) & Limes (Citrus Aurantifolia/Latifolia) Fresh / Dried

No.	State	Value USD Mn	Destination
	All India	8.93	UAE, Nepal, Bhutan (+17)
1	Maharashtra	4.21	UAE, Qatar, Nepal (+9)
2	Uttar Pradesh	1.53	Nepal, UAE
3	Karnataka	1.32	Nepal, UAE, Oman (+5)



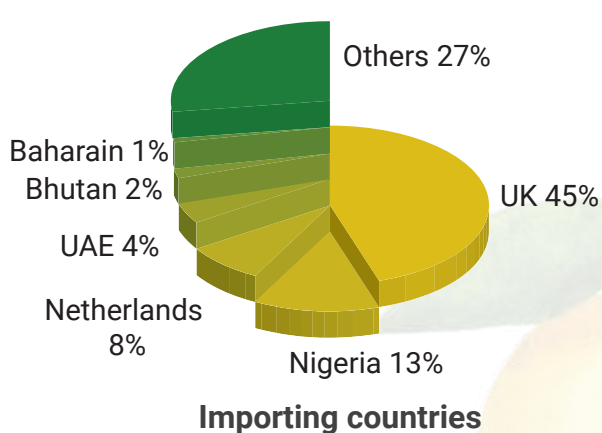
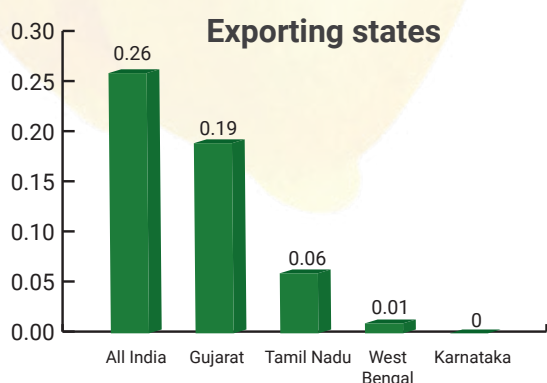
08059000: Other Citrus Fruit Fresh/Dried.

No.	State	Value USD Mn	Destination
	All India	3.49	Bangladesh, Nepal, UK (+23)
1	Punjab	0.94	Bangladesh
2	West Bengal	0.89	Bangladesh, Nepal, Qatar (+4)
3	Maharashtra	0.80	Bangladesh, UK, Canada (+17)
4	Karnataka	0.50	Bangladesh, Canada, Maldives (+5)



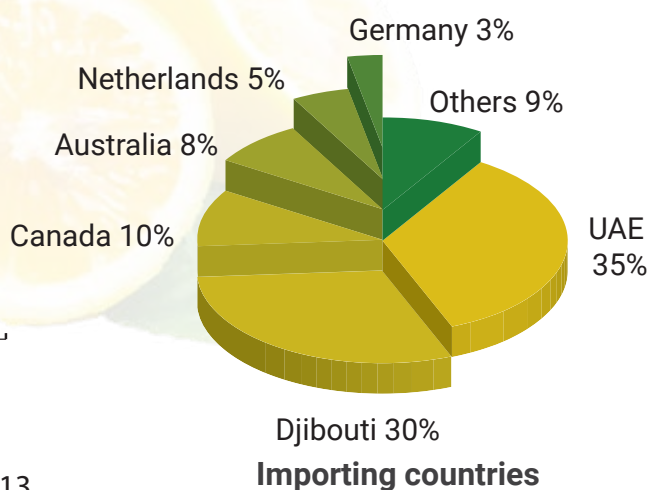
20089912: Lemon Squash.

No.	State	Value USD Mn	Destination
	All India	0.26	UK, Nigeria, Netherlands (+20)
1	Gujarat	0.19	UK, Netherlands, UAE (+7)
2	Tamil Nadu	0.06	Nigeria, Singapore, Oman
3	West Bengal	0.01	Bhutan, UK, Sweden (+2)
4	Karnataka	Nil	NA



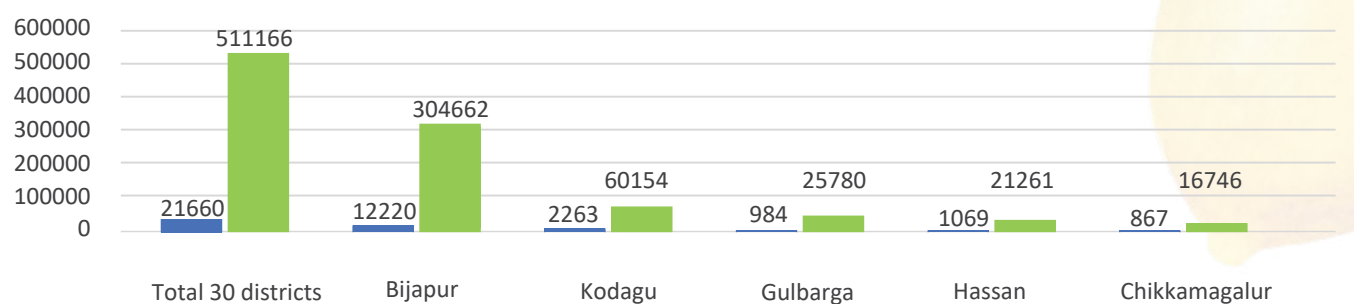
22021020: Lemonade.

No.	State	Value USD Mn	Destination
	All India	0.10	UAE, Djibouti, Canada (+14)
1	Delhi	0.04	UAE, Netherlands, Germany (+4)
2	Maharashtra	0.01	Djibouti, UAE, New Zealand (+4)
3	Gujarat	0.01	Canada, Australia, UK, Tanzania
4	Karnataka	Nil	NA



Production data of Lemon in Karnataka (2021-22): Top 5 districts

Sl.no	District	Area (ha)	Production (m. Ton)
1	Vijayapura	12220	304662
2	Kodagu	2263	60154
3	Kalaburgi	984	25780
4	Hassan	984	21261
5	Chikkamagaluru	867	16746
6	Total 30 districts	21660	511166



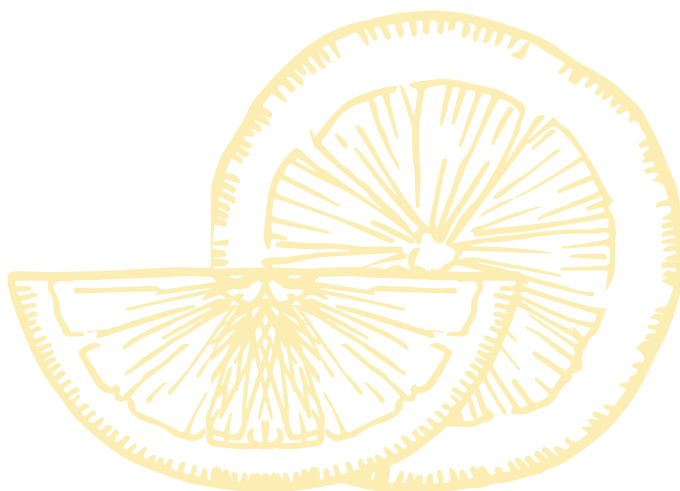
*Source: Lime Board, Horticulture Dept.

FPO's registered and working on Mangoes in Karnataka

1. Basavana Bagewadi Horticulture Farmer Producer Company Ltd, Kolar – Banana, Onion, Grapes and Lime.
2. Shanteshwar Horticulture Farmers Company, Indi – Lime, grapes, Pomegranate and vegetables.
3. Bijapur Horticulture Farmer Producer Company Ltd., Takkalaki – Grapes.
4. Sindagi Horticulture Farmer Producer Company Ltd, Devar Hipparagi – Lime, Grapes and Pomegranate.

Processing technologies available

- Hand operated lemon cutting machine -
<https://cftri.res.in/technologies/FM/hol.pdf>
- Clear Lime-Lemon flavour blend for soft drink manufacture -
<https://cftri.res.in/technologies/BVP/cll.pdf>
- Dehydrated whole lime - <https://cftri.res.in/technologies/FVP/dwl.pdf>
- Carbonated fruit beverages from selected fruits (mango, grapes, lime, orange)-
<https://cftri.res.in/technologies/FVP/cbjuice.pdf>



SPS -TBT Standards for Mangoes

- ▶ Each consignment sample must be visually inspected according to official phytosanitary procedures pertaining to importing countries.
- ▶ Fruits free from visually detectable regulated pests, marks and spots.
- ▶ The fruit may have the calyx attached but not the stem, leaves or other plant parts.
- ▶ Hybrids between the species can be imported but only when each species is an approved commodity, i.e., each species must have an IHS. The hybrid should meet the requirements specified in the IHS for each species in the hybrid. For Example, Orangelo (Citrus paradisi x Citrus sinensis) should meet the requirements specified on the standard for grapefruit (Citrus paradisi) and the standard for orange (Citrus sinensis). The exception to this is when specific hybrid species have been assessed and approved for import. In such cases, the approved hybrid species will be listed in an IHS.
- ▶ Some processed fresh commodities (commercially manufactured and processed fruits and vegetables of specific commercial brands that have undergone pre-export assessment by the Agricultural and Processed Food Products Export Development Authority (APEDA) and they can be found on the APEDA website in “Processed” Fresh Commodities.
- ▶ Frozen, cooked, pickled, pureed or dried commodities for human consumption must be covered by the IHS.
- ▶ Pests, pesticide and insecticide levels for each country are well indicated as ‘maximum residue limits (MRLs) for Pesticide and other chemicals in the food products’ in Indian trade portal for exporter’s reference.
- ▶ Regulated pest list, bacteria, insects, mites, fungus, viruses and viroids along with their treatment requirements for each importing country are explained in detail on World Trade Organisation (WTO) website. (www.wto.org)

Action Plan

- Import best agricultural practices and bring in global varieties germplasm to improvise local varieties by agricultural and horticultural universities.
- Focus on quality development and yield improvement in existing lemon varieties of India.
- Orientation and Facilitation must be prioritised to bridge the gap between research institutes, academia, industries, and farmers to facilitate exports.
- A local APEDA testing labs and pack houses helps growers to understand quality standards and tap international markets.
- Encourage first generation entrepreneurs and start-ups to establish themselves in processing sector with effective hand holding. These industries can produce value added products thereby diverting the surplus produce and avoiding financial losses.
- State and Central governments must sponsor financial schemes/ subsidies to aid capacity building in setting up of industries and encourage skill development programmes for the youth.
- Create awareness about the new products which have international market and transfer knowledge about required technology through CFTRI and IIFPT or any other institutes, to produce value added products.

Opportunities in Lemon and related products

- Though fresh lemon has a huge international market, only 20-25% meets the said standards hence rest gets into local market. The surplus produce can be converted into value added products such as lemon concentrate, lemonade, citrate of lime and desiccated lime powder.
- These value-added products provide employment to a larger group locally and also help the industries to establish themselves in exports, as there is lot of demand in these sectors yet unexplored by Indian growers.
- Setting up of a dedicated testing lab, either stationary or mobile.
- Dedicated pack house in North Karnataka is the need of the hour.
- Processing industries for desiccated powder, taste makers, lemon concentrate will easily fetch revenues from local as well as international market.

Responsibility Matrix

Sl.no	Actions	Responsibility
1	Research on best global varieties	Agricultural and Horticultural Universities, Karnataka.
2	Delegation of progressive farmers to Spain, California or USA.	Lime Board, Karnataka.
3	Farmer Sensitization program on available opportunities	Horticultural department and Lime Board, Karnataka.
4	Inviting Exporters/entrepreneurs to Investors meet for contract farming and processing industries	Lime Board, Karnataka and DIC Districts.
5	Farmer database and aggregation of the produce.	Lime Board, Horticultural department and related FPOs
6	Processing technology and Value Addition Handholding.	Lime Board through CFTRI, Mysore and IIFPT, Thanjavur.
7	Export training, orientation, and Market Intelligence	VTPC Karnataka

Regulators and Service providers

No.	Organization	Service	Contact
1.	APEDA	RCMC, Market survey and assistance.	1st Floor, Beeja Bhavan, Bellary Rd, Hebbal, Bengaluru - 560024. E: apedabl@apeda.gov.in
2.	KAPPEC	PMFME Scheme	17, Richmond Rd, Shanthala Nagar, Richmond Town, Bengaluru - 560025. E: kappec1996@gmail.com
3.	DGFT	IEC, Customs and ICEGATE	6th floor, Kendriya Sadan, C & E Wing, 17th main, Koramangala 2nd Block, Koramangala, Bengaluru - 560034 E: bangalore-dgft@nic.in
4.	Plant Quarantine	Phyto Sanitary certification (SPS)	Hebbal-Boopasandra Road HA Farm Post, Bengaluru - 560024. E: dd-pqfsb-ka@nic.in
5.	BIAL Cool Port	Freight and Flight	KIAL Road, Devanahalli, Bengaluru -560300 W: http://www.aisats.in
6.	Lime Board		Karnataka State Lime Development Board, 3rd Floor, Mini Vidhanasoudha, Indi, Vijayapura E: mdksldb@gmail.com